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Orchestration is carried out using Azure Logic Apps, these will call various data factory pipelines. There are two main types

These data factory pipelines are configured to write to the Unprocessed folder (see structure below).

currently only OPFS and Ctrack SFTP server source systems are supported. These will be extended in the future to cover other sources, such as ODBC database connection.

Some pipelines will do a binary copy of data (i.e. lift and shift the file without processing), others such as database connection will read the data delta and write it out in a supported file format such as CSV, or in the future, JSON.

Raw to Base (contains tag R2B in pipeline name).

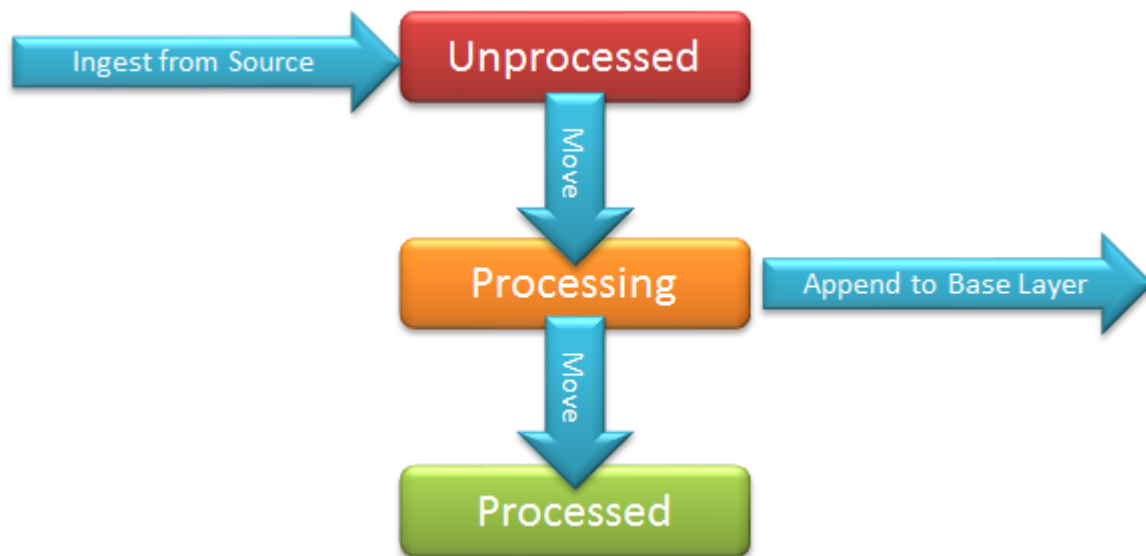
Thereafter, the data will be processed, by calling a data factory pipeline which in turn calls the Databricks Ingestion Pattern, to undertake the following steps (also illustrated below):

1. Move the Raw data files into the Processing folder
2. Cleanse the data and append it to the Base Layer Databricks Delta file structure*
3. Move the processed Raw data file into the Processed folder structure (see blow). The Raw file written here are unchanged and identical to those received in the Unprocessed folder.

*Although the pattern is capable of handling schema drift, a standard has been adopted to always specify a fixed schema up-front via the AWBI System entity metadata (see support pages for examples).

Raw

Illustration of dataflow through the Ingestion folders:



- Structure and pattern to be defined - This would be potentially for unstructured file types such as images, sound files etc which we would not be able to convert into our Delta base format.

Folder Structure - Unprocessed

The data factory ingestion pipelines is configured to store the all raw data files in folders as follows :

/root/raw/unprocessed{sourceSystem}/{fileType}/{entityname}_{processingtimestamp}.{extension}

Where:

- *{Source System}* - e.g. ctrack, filestore, ei. The short name of the source system from where data has been ingested from
- *{fileType}* - e.g. csv, excel, shape etc. The file type as define in the supported formats define above.
- *{entity}* - Physical Name of source Table or File.
- *{processingtimestamp}* - time stamp of when the file was picked up by ADF

Example. CTrack file AW_GeoZones:

/root/raw/unprocessed/ctrack/csv/aw_geozones_20170801040332.csv/

Folder Structure - Processing

When Databricks Ingestion Pattern is run, for the particular *{File Type}*, the files are moved from the Unprocessed folder to the Processing folder structure. This folder structure is the same as Raw.

Example. CTrack file AW_GeoZones:

/root/raw/**processing**/ctrack/csv/aw_geozones_20170801040332.csv/

Folder Structure - Processed

Once the Databricks Ingestion Pattern has completed appending the cleansed Raw files into the Base Layer it moves them into the Processed folder structure. This folder structure is the same as Raw.:

Example. CTrack file AW_GeoZones:

/root/raw/**processed**/ctrack/csv/aw_geozones_20170801040332.csv/

Base

When writing the cleansed data out to the base layer the storage technology is now ADLS Gen 2 (not BLOB storage like RAW). The file format here is always 'Databricks Delta' so the file-type is no longer required. The filename has also been prefixed with the source system name, and the file extension has been removed as it is no longer relevant, as follows:

`/root/base/{sourceSystem}/{sourceSystem}{entityname}_{processingtimestamp}`

Example. CTrack file AW_GeoZones:

`/root/base/ctrack/aw_geozones_20170801040332/`

Data Cleansing

The main value add of the "Base" layer is data cleansing, data should be consistently cleansed and ready for data processing. The data cleansing actions that occur within the notebook are described below;

Trimming spaces from left and the right of text

Replacing string attributes that contain null with '?'

Dropping duplicate rows

Bad Data

Bad data is not currently handled. A load option of dropmalformed rows is available but this should be used cautiously

Support

See the Wiki support pages for examples of configuring and maintenance of various generic ingestion scenarios.